

10.1: Approximations with Taylor Polynomials.

HW problems 21-22 and 33-36. *Approximate the given quantities using Taylor polynomials with $n = 3$, and compute the absolute error in the approximation assuming the exact value is given by a calculator.*

1. $e^{0.2}$

2. $\ln(1.01)$

3. $\sin\left(\frac{\pi + .02}{2}\right)$ use center $\frac{\pi}{2}$.

10.2: Find the Interval of Convergence.

HW problems 9-10 and 21-24. *Determine the radius of convergence of the following power series. The test the endpoints to determine the interval of convergence.*

4. $\sum \left(\frac{3x}{2}\right)^k$

5. $\sum \frac{k^2 x^{2k}}{k!}$ (use the Ratio test)

HW problems 21-24. Find the power series representations centered at 0 for the following functions using known power series. Give the interval of convergence for the resulting series.

6. $f(x) = \frac{1}{1-2x}$

7. $f(x) = \frac{2x^3}{3-3x}$

11.1: Working with Parametric Equations.

HW problems 7-8. Plot the parametric points for the given values of t , and eliminate the parameter to an equation in terms of x and y .

8. $x = -t + 6$, $y = 3t - 3$; for $t = -5, -1, 0, 1, 5$

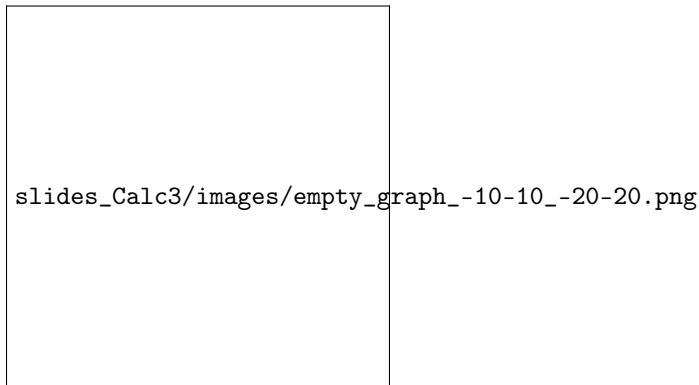


Figure 0.1: Use the graph to complete the preceding problem.

HW problems 23-24 and 35-36. Give a set of parametric equations that describes the curves.

9. The complete curve $x = y^2 - 2y$.

10. A man jogs clockwise at a constant speed around a circular track of radius 50 yards completing one lap in 2.5 minutes.

HW problems 45-48. Determine dy/dx in terms of t and evaluate it at the given value of t .

11. $x = 2t$, $y = t^3$; $t = -1$

11.2: Working with Polar Coordinates.

HW problems 9-11. Graph the points in the following polar coordinates.

12. $\left(1.5, \frac{7\pi}{4}\right)$, $\left(-1.5, \frac{7\pi}{4}\right)$, $\left(1.5, -\frac{7\pi}{4}\right)$, $\left(1.5, \frac{7\pi}{4} + 2\pi\right)$, $\left(-1.5, \frac{7\pi}{4} + \pi\right)$

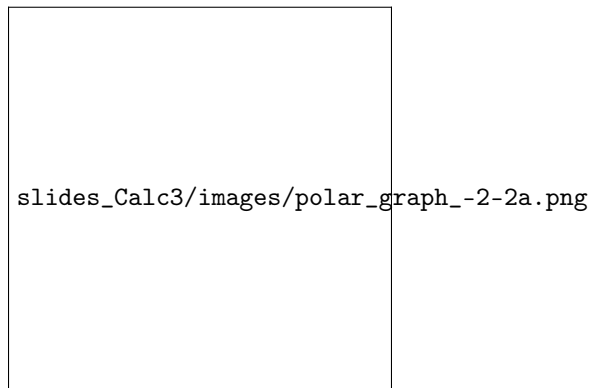


Figure 0.2: Use this graph to plot the polar coordinates

HW problems 15-17. Express the following polar coordinates in Cartesian coordinates.

13. $\left(2, \frac{7\pi}{4}\right)$

HW problems 21-24. Express the following Cartesian coordinates in polar coordinates in at least two different ways.

14. $(4, \sqrt{3})$

HW problems 45. Mark the points on the polar graph that corresponds to the points shown on the Cartesian graph.

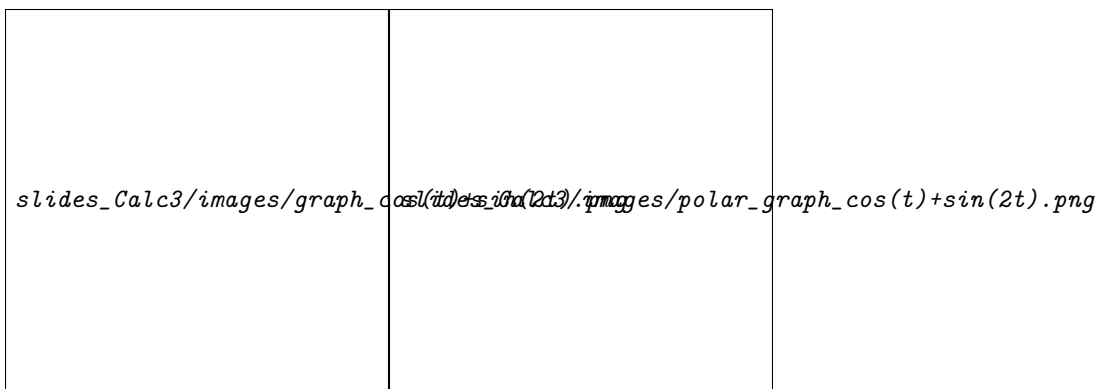


Figure 0.3: $r = \cos(\theta) + \sin(2\theta)$

15. Mark the following points: A, B, C, D, E, G, H, I, J

11.3: Finding Derivatives and Integral in Polar Coordinates.

HW problems 5-11. Find the slope of the line tangent to the following polar curves at the given points.

16. $1 + \cos \theta$; $\left(1 + \frac{\sqrt{3}}{2}, \frac{\pi}{6}\right)$, $(1, 2\pi)$

HW problems 21-25 (odd). Find the area of the region.

17. The region inside the limaçon $r = 2 + \cos \theta$.

18. The region inside one leaf of the rose $r = \cos 5\theta$.